Solution:

Since every positive number is greater than every negative number,

we conclude that $\frac{5}{17} > \frac{-10}{19}$.

Type 2 : Comparing two rational numbers represented by two fractions with same denominators

Example 1.4

Compare $\frac{1}{3}$ and $\frac{4}{3}$.

Solution:

Since the denominators are the same, just compare the numerators.

Since $1 < 4$, we conclude that $\frac{1}{3} < \frac{4}{3}$.

Type 3 : Comparing two rational numbers represented by two fractions with different denominators

Example 1.5

Compare $\frac{3}{4}$ and $\frac{5}{6}$.

Solution:

The LCM of the denominators is 12 (Find it!). Consider for each rational number an equivalent rational number with the LCM 12 as denominator.

We get $\frac{3}{4} = \frac{9}{12}$ and $\frac{5}{6} = \frac{10}{12}$, which become like fractions now.

Here, $\frac{9}{12} < \frac{10}{12}$. Hence, we conclude that $\frac{3}{4} < \frac{5}{6}$.

Type 4 : Comparing two rational numbers that are not in standard form

Example 1.6

Compare $\frac{9}{-4}$ and $\frac{-2}{3}$.

Solution:

The number $\frac{9}{-4}$ is not in standard form. First put it in the standard form.

$$\frac{9}{-4} = \frac{9}{-4} \times \frac{-1}{-1} \text{(to get a positive denominator)} = \frac{-9}{4}$$

Now, we shall compare the fractions $\frac{-9}{4}$ and $\frac{-2}{3}$. We find that these two fractions are unlike fractions. To make them as like fractions, we make use of their LCM, which is 12.

We can now compare their equivalent fractions $\frac{-9}{4} = \frac{-27}{12}$ and $\frac{-2}{3} = \frac{-8}{12}$ (How?). We find that the denominators are the same and so just comparing the numerators –27 and –8 are enough.

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